

Covalent and ionic substances have different properties

Melting and boiling points

Ionic substances have very high melting points and boiling points. This is because there are very strong electrostatic forces holding the ions together.

Covalent substances made from simple molecules have low melting points and boiling points because, although the forces holding the molecules together are strong, the forces between the molecules – intermolecular forces – are weak. This means that only a small amount of energy is needed to overcome these forces in order to melt or boil them.

Questions

Use this table to help answer questions 5–9.

Substance	Melting point in °C	Boiling point in °C
sodium chloride	801	1413
methane	–182	–161
magnesium chloride	714	1412
ammonia	–77	–34
calcium oxide	2613	2850
chlorine	–101	–34.6
water	0	100

- 5 Is magnesium chloride an ionic compound or a simple molecule with covalent bonds? Give reasons for your answer.
- 6 Is ammonia an ionic compound or a simple molecule with covalent bonds? Give reasons for your answer.
- 7 Is ammonia a solid, liquid or gas at room temperature?
- 8 Why do magnesium chloride and calcium oxide have high melting points?
- 9 Why do methane and chlorine have low melting points?

- 10 Look at this photograph of copper sulfate crystals.

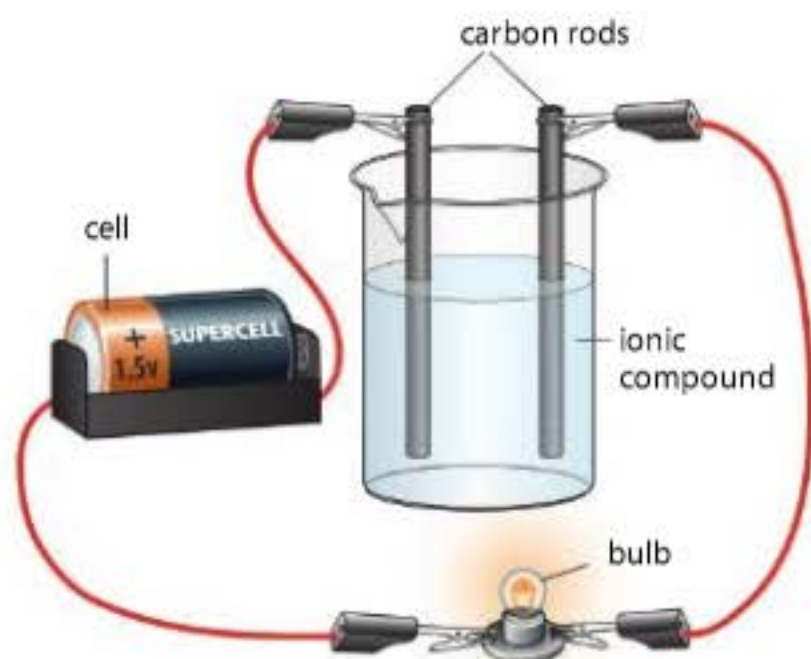


What sort of chemical bonds does this substance have?
Give a reason for your answer.

- 11 A substance has a melting point of $3078\text{ }^{\circ}\text{C}$ and a boiling point of $4300\text{ }^{\circ}\text{C}$. What sort of chemical bonds does it have? Give a reason for your answer.
- 12 Silicon dioxide is a compound formed from two non-metals, silicon and oxygen. It is the chemical name for sand. It is a hard substance with a melting point of $1610\text{ }^{\circ}\text{C}$. Which properties of silicon dioxide suggest that it has a giant covalent structure?

Conducting electricity

Ionic compounds will conduct electricity if they are dissolved in water or if they are melted to form a liquid.



Ionic compounds can conduct electricity because the ions have an electrical charge. The ions must be free to move about and carry the electrical charge. Covalent substances made from simple molecules do not conduct electricity.

Think like a scientist

Ionic compounds conducting electricity

You will need:

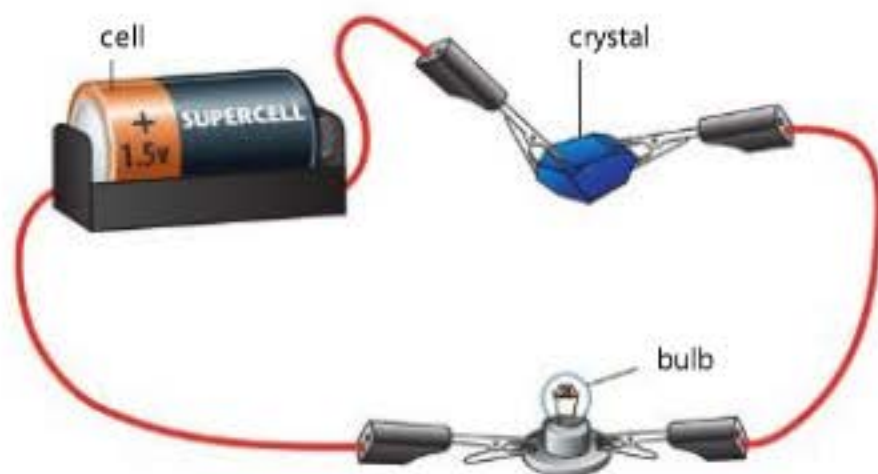
- safety glasses, surgical gloves, electrical wires, lamp, cell, carbon electrodes, beaker, scalpel or knife, selection of crystals of ionic compounds such as copper sulfate and sodium chloride, selection of crystals of ionic compounds, such as copper sulfate and sodium chloride.

Safety

Some crystals and solutions such as copper sulfate are irritants. Take care when using these and avoid touching them. Wear surgical gloves and safety glasses.

Method

- 1 Set up your apparatus as shown in the diagram on the previous page, but without the solution. Test to check that the bulb is working before you start.
- 2 Place a solution of one of your ionic compounds in a beaker.
- 3 Add your beaker containing the ionic compound to your circuit, but do not connect the final link until you have checked your circuit carefully.
- 4 Complete the circuit.
- 5 Repeat this for all the solutions you are using.
- 6 Record your results.
- 7 Place one of the crystals in position in the circuit and then complete the circuit. Take care that the connections on the crystal are not touching.



- 8 Repeat this for all the crystals you are using.
- 9 Record your results.

Questions

- 1 Did all your ionic solutions conduct electricity?
- 2 Explain your answer.

Continued**Questions**

- 3 Did all the ionic crystals conduct electricity?
- 4 Explain your answer.
- 5 Predict, with reasons, what would happen if you used a covalent substance in place of an ionic crystal or solution.

Summary of the properties of ionic and covalent substances

Ionic substances	Covalent substances
Compounds made from ions form giant lattices. These ionic compounds have very high melting points and boiling points because the forces between the ions are so strong.	Substances made from simple molecules have low melting points and boiling points because there are only weak forces between the molecules. Atoms that share electrons can form giant structures called macromolecules. These have very high melting points because the atoms are joined by strong covalent bonds.
When ionic compounds are dissolved in water or melted, they can conduct electricity because the ions are free to move about and carry the electrical charge.	Simple covalent molecules do not conduct electricity.

Summary checklist

- ☐ I can explain how giant structures are formed.
- ☐ I can compare the properties of ionic and covalent substances.
- ☐ I can relate the structures of these substances to their properties.